6 Theory: stability to subspace interference

We now investigate the impact of interfering subspaces. Consider the almost-separable limit, modelling interference as a random infinitesimal perturbation of the vectors $\{q, k_t, m_t\}$. Let ϵ-symbols denote perturbations such that $\epsilon^{\Delta x(q)} \to \frac{\partial \Delta x}{\partial q}\epsilon^q$ for $\epsilon^q \to 0$ is the change of Δx induced by ϵ^q. We consider (i) the sparse limit, in which the attention is concentrated entirely on a single embedding, and (ii) the isotropic limit, in which it is distributed evenly among embeddings. We are particularly interested in the sparse case, since this highly directed flow of information is used by match&pass, although semantic separation can also be used by non-sparse heads.

[Definition] Sparse attention: the low-temperature limit $a_t \approx \delta_{tt^*}$ and $\Delta x = m_{t^*}$, where δ is the Kronecker delta. This occurs when there is a large difference between the top two scores: $t^* = \mathrm{argmax}_t w_t$ and $w_{t^*} - \max_{t \neq t^*} w_t \gg 1$. **Isotropic attention:** the high-temperature limit $a_t = \frac{1}{T}$ and $\Delta x = \langle m_t \rangle_t$. This occurs when w_t is constant, requiring $q = 0$ or constant k_t.

Stability of attention updates to perturbations on q, k, v *[Proofs in appendix F]*

Theorem 4. *Consider independent infinitesimal perturbations on queries $\epsilon^q \in \mathbb{R}^{N_{qkv}}$, keys $\epsilon^k_t \in \mathbb{R}^{N_{qkv}}$, and messages $\epsilon^m_t \in \mathbb{R}^{N_{qkv}}$. These propagate onto $\Delta x = \sum_t a_t m_t$ as*

$$\epsilon^{\Delta x(q)} \xrightarrow[\epsilon^q \to 0]{\text{perturb q}} \mathop{\mathbb{E}}_{a_t}\left[m_t \tilde{k}_t^T\right]\epsilon^q \qquad \tilde{k}_t \triangleq k_t - \mathop{\mathbb{E}}_{a_t}\left[k_t\right] \tag{7}$$

$$\epsilon^{\Delta x(k)} \xrightarrow[\epsilon^k_t \to 0]{\text{perturb k}} \mathop{\mathbb{E}}_{a_t}\left[\tilde{m}_t {\epsilon^k_t}^T\right] q \qquad \tilde{m}_t \triangleq m_t - \mathop{\mathbb{E}}_{a_t}\left[m_t\right] \tag{8}$$

$$\epsilon^{\Delta x(m)} \xrightarrow[\epsilon^m_t \to 0]{\text{perturb m}} \mathop{\mathbb{E}}_{a_t}\left[\epsilon^m_t\right] \tag{9}$$

where $\tilde{z}_t$ is the value of z_t measured from the attention-weighted centroid $\mathbb{E}_{a_t}[z_t] = \sum_t a_t z_t$.

Theorem 5. *For sparse attention:*

$$\epsilon^{\Delta x(q)} \xrightarrow[\epsilon^q \to 0]{\text{perturb q}} 0 \qquad \epsilon^{\Delta x(k)} \xrightarrow[\epsilon^k_t \to 0]{\text{perturb k}} 0 \qquad \epsilon^{\Delta x(m)} \xrightarrow[\epsilon^m_t \to 0]{\text{perturb m}} \epsilon^m_{t^*} \tag{10}$$

i.e. the message is stable with respect to small interference in the queries and keys. Interference in the selected value is linearly transferred onto the message.

Theorem 6. *For isotropic attention:*

$$\epsilon^{\Delta x(q)} \xrightarrow[\epsilon^q \to 0]{\text{perturb q}} \langle m_t \tilde{k}_t^T \rangle_t \epsilon^q \qquad \epsilon^{\Delta x(k)} \xrightarrow[\epsilon^k_t \to 0]{\text{perturb k}} \langle \tilde{m}_t {\epsilon^k_t}^T \rangle_t\, q \qquad \epsilon^{\Delta x(m)} \xrightarrow[\epsilon^m_t \to 0]{\text{perturb m}} \langle \epsilon^m_t \rangle_t \tag{11}$$

N.B. isotropy requires $k_t = const$ or $q = 0$. Lemma 1: the update is stable to noisy q when $k_t = const$, or when $m_t \perp k_t$ (c.f. keys and messages from independent subspaces). Lemma 2: the update is stable to noisy k_t when $q = 0$, or when $m_t \perp \epsilon^k_t$. Lemma 3: the update is stable to noisy m_t when $\langle \epsilon^m_t \rangle_t = 0$. Other cases propagate linearly.

The stability of sparse attention is because `softmax` becomes an `argmax` for low-temperature heads, which is only sensitive to the order of w_t. However, this introduces a different vulnerability when perturbations cause the order of w_t to change, as the attention distribution undergoes a phase transition to select a different token. We call this *circuit collapse*. For example, the induction circuit collapses when the operation *attend to the previous token* attends to any other token because of interference.

[Definition] Circuit collapse: spontaneous phase transition in which a sparse attention distribution selects a different token due to noise on $\{q, k_t\}$. Let $\epsilon^w_t = k_t^T \epsilon^q + q^T \epsilon^k_t + \mathcal{O}({\epsilon^q}^T \epsilon^k_t)$ be perturbations on w_t that result from ϵ^q and ϵ^k_t. Circuit collapse occurs when there exists a $t \neq t^*$ for which $w_{t^*} - w_t < \epsilon^w_t - \epsilon^w_{t^*}$.

We now study the L_2-norm interference that we expect to be induced by `Pre-Norm` when semantic separability is violated. This is characterised by perturbations that are parallel to their corresponding vector. Theorem 7 shows the conditions under which we expect circuit collapse to occur.

Stability of attention updates to scaling of q, k, v *[Proofs in appendix F]*

Theorem 7. *Sensitivity of sparse attention to multiplicative perturbations $\epsilon^q = \kappa^q q$ and $\epsilon^k = \kappa^k_t k_t$ with $\kappa^q, \kappa^k_t \ll 1$. Circuit collapse occurs when $\exists\, t \neq t^*$ for which:*

$$\frac{w_{t^*}}{w_t} \begin{cases} < \lambda_w & \text{if } w_t\left(1+\kappa^q+\kappa^k_{t^*}\right) > 0 \\ > \lambda_w & \text{otherwise} \end{cases} \qquad \lambda_w \triangleq \frac{1+\kappa^q+\kappa^k_t}{1+\kappa^q+\kappa^k_{t^*}} \tag{12}$$

where temperature cancels in the fraction. ***Attention is fully stable above the critical transition point*** *λ_w (c.f. $w_t\left(1+\kappa^q+\kappa^k_{t^*}\right) > 0$). We see that query perturbations alone are insufficient, as they result in $\lambda_w = 1$. Lemma: consider the special case when all keys have similar length $k_t \approx const$, the attended token has $\theta_{t^*} \approx 0$, the keys are far-from-orthogonal s.t. $\theta_t \ll 1$, and $\kappa^q \approx 0$. Using $w_t \triangleq |q||k_t| \cos\theta_t$, circuit collapse occurs when $\exists\, t \neq t^*$ for which:*

$$\frac{1}{2}\theta_t^2 \lesssim \kappa^k_t - \kappa^k_{t^*} \qquad \text{if } w_t\left(1+\kappa^k_{t^*}\right) > 0\text{, otherwise reverse} \tag{13}$$

i.e. stability requires either well-separated keys s.t. $\theta_t \gg 0$, or small perturbations $\kappa_t - \kappa^*_t \ll 1$.

Theorem 8. *Sensitivity of isotropic attention to multiplicative perturbations. Say $\epsilon^k = \kappa^k_t k_t$ with $\kappa^k_t \ll 1$ where $\{\kappa_t\}$ have comparable amplitudes. Then*

$$\epsilon^{\Delta x(k)} \approx \begin{cases} 0 & \text{if } \kappa_t \text{ independent of } \tilde{m}_t\text{, by symmetry} \\ 0 & \text{if } \kappa_t \equiv \kappa \text{ for constant } \kappa \\ 0 & \text{if } q = 0 \\ w\langle \tilde{m}_t \kappa^k_t \rangle_t & \text{otherwise} \end{cases} \tag{14}$$

7 Experimental results

We now use experiments to empirically probe (i) the real-world embedding structure, and (ii) the sensitivity to artificial noise on the {query, key, value} L_2-norms. Whilst this does not directly observe real-world interference, it constrains the effect importance.

We consider a base-10 integer-addition task with a question-answer structure, and train for next-token prediction. We use a decoder architecture, common for state-of-the-art language models, with 10 layers, per-character tokenisation, and begin `[` and end `]` tokens. In the output, we mask `*` tokens that precede the answer. For example, the first training sequence has input `[453+16+17-N846=1332` and output `**********1332]`. We compare two models that use `Pre-Norm` and `QKV-Norm` respectively. Appendices A-C provide a full experimental setup and supplementary plots. In this section we make all plots using an in-distribution test set that is expected to have some overlap with the training set, bounded at $\ll 20\%$.

We choose this task because it emphasises contextual reasoning in a small-scale setting, is configurable for complexity, and allows us to define meaningful out-of-distribution test sets. The `Pre-Norm` (`QKV-Norm`) model achieves an in-distribution per-token accuracy of 91.4% (91.0%), dropping to 87.5% (82.5%) when generalising out-of-distribution to intermediate complexity, and 66.7% (46.8%) for increased complexity. Statistical uncertainties are below 0.1%. The in-distribution performance is comparable, but `QKV-Norm` generalises worse in this task, implying it has learned less task-appropriate solutions. Appendix D shows additional comparisons suggesting that the `Pre-Norm` and `QKV-Norm` models behave differently, supporting the observations of [18–20].

Embedding structure

Our theory predicts that `Pre-Norm` attention is stable with respect to information in non-attended subspaces if all input embeddings have similar L_2-norms, whereas `QKV-Norm` imposes no norm constraint. We seek to experimentally bound the degree to which this structure is learned in practice.

We do this by plotting *the spread of norms with respect to their median*. A confounding effect is that the norms may differ for (i) embeddings attended to by different heads, (ii) the same head acting in different contexts, and (iii) embeddings that are never attended. We therefore measure the ratio *per-head*, and weight each embedding by its assigned attention. We remove the begin-sequence token from consideration. Figure 1 shows the resulting spread for all attention layers. On the LHS, we see that 90% of the distribution is contained within an interval of $\pm 20\%$ when using `Pre-Norm`. On the RHS, we see that `QKV-Norm` allows a much wider spread. This is consistent with our theory, and experimentally bounds the representation effect on `Pre-Norm` to $\lesssim 20\%$ in this model. Supplementary Figures 15-16 show consistent results for two model variations, although we note that `Pre-Norm` and `QKV-Norm` are more comparable for the variation labelled `Alternate`.

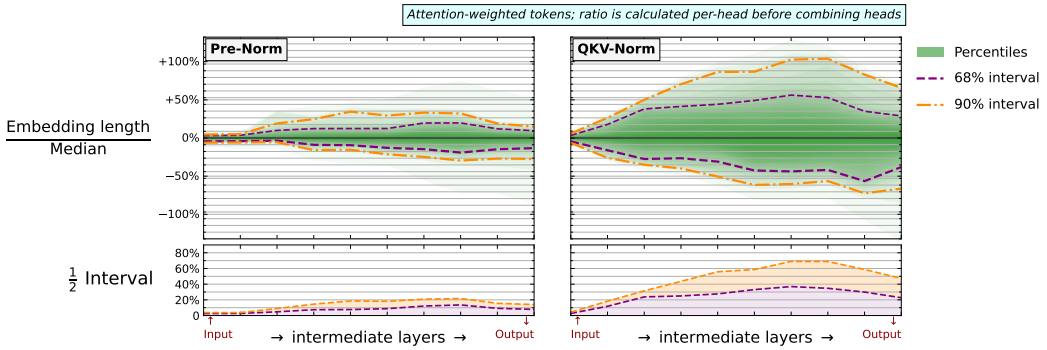

Figure 1: Spread of embedding L_2-norms experienced by attention heads at increasing model depth, excluding the `[` token. For `Pre-Norm`, 90% of the spread is observed within an interval of $\pm 20\%$. Supplementary Figure 11 shows the distributions used to make this plot. Supplementary Figures 15-16 replicate the analysis for two model variations.

Model stability with simulated interference

Section 6 theoretically modelled the semantic interference induced by `Pre-Norm` as a random perturbation on the norms of $\{q, k_t, m_t\}$. To estimate the real-world sensitivity to such an effect, we artificially introduce uncorrelated uniform noise onto these norms inside our trained `Pre-Norm` model. Even though Gaussian noise is expected in the large-N_s limit, we use uniform noise to avoid outliers. Figure 2 shows the evolution of in-distribution per-token accuracy with increasing RMS. On the LHS, we see that performance falls by $\gtrsim 10\%$ at only a 1% noise level. We also show the trend excluding the end-sequence token, which contributes a significant fraction of the metric. On the RHS, we introduce $\{q, k_t\}$ noise only to sparse heads (when $\max_t a_t \geq 95\%$) and non-sparse heads (when $\max_t a_t < 70\%$). We see that the model is stable with respect to $\%$-scale noise on sparse-attention, and this regime is dominated by the non-sparse case.

Figure 2 (right) is consistent with the stability predictions of Theorems 5-6. However, it may also be explained if non-sparse distributions are simply more important to the model. This could be caused by non-sparse distributions being more common, as well as depth-dependence. This is because artificial noise is applied to all layers during the forward pass, therefore later layers are perturbed by both the noise component *and* the shifting of their inputs due to previous layers, which is expected to compound with depth. We are interested in capturing this effect, however it may increase the importance of early layers. See Figure 26 for a visualisation of the observed attention maps.

Circuit collapse

Figure 3 shows the probability that our artificial noise causes the circuit collapse phenomenon as defined in section 6. In this experiment, we add noise to every layer independently. This prevents the confounding effect of shifting inputs due to noise in previous layers. We observe that 1% of sparse attention distributions collapse when they experience noise at a level of 11%. This reduces to 7.5% and 5.5% for the two model variations shown in Appendix C.